

Q1-Style Peer Review of PoxCSAF-Net

Manuscript: PoxCSAF-Net: Cross-Scale Attention Fusion for Privacy-Preserving Federated Classification of Mpox and Visually Similar Skin Lesions

Reviewer Recommendation: Major Revision

The manuscript has a potentially publishable core. The combination of multiclass Mpox classification, cross-dataset deduplication, federated learning, external validation, calibration, DP-SGD, poisoning analysis, and personalization gives it considerably more depth than a straightforward image-classification paper. The proposed architecture also combines ConvNeXtV2-Tiny with ECA, CBAM, and a cross-scale attention head.

However, several methodological claims need substantial tightening before the manuscript reaches the standard expected of a strong Q1 journal in medical AI, biomedical imaging, or federated learning.

~~Major Comment 1. The privacy-preserving claim is currently too strong~~

The title describes the method as privacy-preserving, yet the manuscript states that differential-privacy accounting is scoped to each client-local update within each communication round. It also states that composing this into an end-to-end privacy guarantee is future work.

This means $\epsilon = 8$ or $\epsilon = 12$ is not the privacy budget of the final trained system. Privacy loss composes over repeated access to the same client data.

For a Q1 paper, either compute the cumulative privacy guarantee for the full training process using an appropriate accountant, or remove "privacy-preserving" from the title and consistently describe the method as privacy-aware. In the latter case, DP-SGD should be presented as an exploratory extension rather than a complete privacy guarantee.

~~Major Comment 2. The heterogeneous/non-IID experiment is weaker than the wording suggests~~

The heterogeneous experiment uses client-size weights [0.10, 0.15, 0.30, 0.25, 0.20] while keeping class proportions approximately similar between clients. This primarily represents quantity skew.

The Introduction motivates institutional heterogeneity through differences in class prevalence, image quality, skin tone, lesion stage, lighting, and device type. Those are stronger forms of distribution shift than quantity skew alone.

For Q1-level evidence, add at least one genuine statistical non-IID configuration. A practical minimum would be Dirichlet label skew at multiple α values. A stronger design would also include source-aware or domain-skewed clients where feasible.

~~Major Comment 3. The promised FedAvg versus FedProx comparison is not properly delivered~~

The contribution statement says that both FedAvg and FedProx are evaluated under heterogeneous data conditions, yet the primary classification table compares centralized training, FedProx under homogeneous clients, and FedProx under heterogeneous clients.

That does not provide the clean FedAvg versus FedProx comparison promised by the manuscript. The comparison should use identical folds, initialization, augmentation, local epochs, learning rates, and random seeds so that any difference can reasonably be attributed to the aggregation strategy.

~~Major Comment 4. Dataset leakage control needs more documentation~~

Deduplication before splitting is a strong methodological choice, especially because EMSID incorporates MSID and internet-sourced material. However, saying only that perceptual hashing was applied is insufficient for a rigorous journal paper.

Report the hashing algorithm, hash size, Hamming-distance threshold, exact duplicate count, near-duplicate count, manual verification procedure, and treatment of transformations such as resizing, cropping, compression, and watermarks.

Run a direct deduplication audit between the full development pool and MPox-Vision. Because multiple public MpoX datasets draw from internet sources, external validation can otherwise be compromised by duplicated or derivative images.

Major Comment 5. Patient-level leakage needs to be addressed

The manuscript uses image-level cross-validation, but it is unclear whether multiple images from the same patient, lesion, or clinical case can occur across training and test partitions.

Perceptual hashing cannot guarantee patient-level independence because two photographs from the same patient may show different lesion locations, viewpoints, or disease stages.

If patient or case identifiers are available, splitting should be grouped by patient or case. If they are unavailable, the manuscript should explicitly acknowledge that patient-level separation cannot be guaranteed.

Major Comment 6. The central architectural novelty is not convincingly demonstrated

PoxCSAF-Net combines several established components including ConvNeXtV2, ECA, CBAM, GeM, stochastic depth, focal loss, SWA, and cross-attention. The strongest candidate for architectural novelty is the CSAH design and its integration.

The ablation results appear to show that a simpler ECA/CBAM/GAP configuration can reach a higher mean macro-F1 than the primary model. Selecting the primary architecture because it appears more consistent or has lower variance is not enough on its own.

A stronger CSAH ablation should compare the backbone, a GAP baseline, ECA and CBAM with GAP, multi-scale concatenation with MLP, multi-scale weighted fusion, CSAH without GeM, CSAH with mean pooling, and the full CSAH. Parameters and FLOPs should also be reported.

Major Comment 7. Five-fold Wilcoxon testing is statistically weak

With only five paired observations, Wilcoxon signed-rank testing is severely underpowered. Failure to reject the null hypothesis does not establish equivalence.

Emphasize paired fold-wise differences, mean performance difference, effect size, and bootstrap confidence intervals. The Wilcoxon test can remain as supplementary evidence, but it should not be the main justification for choosing one architecture over another.

~~Major Comment 8. External validation methodology is incomplete~~

MPox-Vision contains Chickenpox, Measles, and Monkeypox but does not contain the Healthy class. The manuscript needs to state exactly how the four-class model is evaluated on this three-class external dataset.

Clarify whether the Healthy probability is retained and counted as an error, or whether probabilities are renormalized over the three available classes.

The manuscript should also fully define any prior-shift correction, including how source and target priors are estimated, whether external labels are used, and whether any parameters are tuned on the external set. If external labels influence the correction, the corrected result should not be presented as a completely untouched external evaluation.

~~Major Comment 9. FedPer is not evaluated as a personalization method should be~~

FedPer maintains client-specific heads, yet the manuscript evaluates the resulting system using a pooled global-style test protocol. That does not directly test whether personalization helps each client.

A more appropriate analysis would compare the global FedProx model and each client-specific FedPer model on that client's own held-out distribution. Report per-client macro-F1 and the improvement or degradation for each client.

This experiment would also become more meaningful under genuine label or domain heterogeneity rather than quantity skew alone.

~~Major Comment 10. The poisoning experiment should be framed as sensitivity analysis~~

Only one client is attacked, and that client is one of the smaller clients under the heterogeneous allocation. This limits its influence on the global update.

The non-monotonic change in F1 as the label-flipping fraction increases suggests that training variability may be comparable to the observed attack effect.

The present experiment is useful as a sensitivity analysis, but it should not be presented as strong evidence of adversarial robustness unless stronger threat models are evaluated.

1 Major Comment 11. Clinical language should remain conservative

The Introduction appropriately states that image-based AI should not replace PCR confirmation. That caution should remain consistent throughout the paper.

The study is based mainly on publicly available images and simulated federated clients rather than a prospective, multi-institution clinical cohort. Claims about real-world hospital deployment, clinical reliability, or practical clinical screening should therefore be carefully qualified.

~~Major Comment 12. No demographic or fairness evaluation~~

The manuscript mentions skin-tone differences as a possible source of client heterogeneity, but it does not perform subgroup evaluation by skin tone or demographics.

If Fitzpatrick type or suitable metadata are unavailable, the manuscript should state this explicitly as a limitation rather than implying that fairness or skin-tone robustness has been established.

Major Comment 13. The manuscript is not submission-ready because template placeholders remain

The first page still contains Frontiers abstract instructions instead of a completed abstract and also retains a placeholder corresponding-author email.

The manuscript also contains unresolved placeholders such as Figure ?? and template material in several end sections. These issues can lead to technical rejection before substantive peer review.

Suggested title revision

Current title:

PoxCSAF-Net: Cross-Scale Attention Fusion for Privacy-Preserving Federated Classification of Mpox and Visually Similar Skin Lesions

Safer title unless full end-to-end DP accounting is added:

PoxCSAF-Net: Cross-Scale Attention Fusion for Federated Classification of Mpox and Visually Similar Skin Lesions

What is already strong

- Four-class classification rather than an easier binary problem
- Deduplication before train/validation/test partitioning
- Five-fold evaluation
- Class-level metrics
- External validation
- Calibration analysis
- Test-time augmentation
- Differential-privacy experiments
- Poisoning sensitivity analysis
- Personalization experiments
- Explainability analysis
- Architecture and training ablations

The principal internal performance is also reasonably strong. The manuscript reports homogeneous federated macro-F1 around 0.925 and heterogeneous FedProx macro-F1 around 0.911. The main weakness is therefore not a lack of experiments. It is that the central methodological claims are broader than the evidence currently supports.

Priority actions before Q1 submission

1. Fix all template problems and write the actual abstract.
2. Run external-dataset deduplication against the full development pool.
3. Add a direct FedAvg versus FedProx comparison under identical conditions.
4. Add at least one genuine label-skew non-IID experiment.
5. Resolve end-to-end DP accounting or remove privacy-preserving from the title.
6. Fully define the external prior-shift correction and three-class evaluation protocol.
7. Strengthen the CSAH ablation because a simpler configuration appears to score higher.
8. Evaluate FedPer per client or move the personalization analysis to a more limited role.
9. Document patient-level or case-level splitting limitations.
10. Tighten clinical and adversarial-robustness claims.

Overall assessment

The manuscript has a credible basis for Q1 peer review after substantial revision. In its current form, the four issues most likely to determine the outcome are privacy accounting, the non-IID experimental design, the external-validation protocol, and the justification of the architectural novelty.

Reviewer recommendation: Major Revision